UNIT 2 · 9 hours

Three-Phase Induction Motor: Principle, Construction and Types



Transformers and Electrical Machines

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- Describe stator and rotor construction
- Distinguish squirrel-cage and wound (slip-ring) rotors
- Explain the principle of operation and slip
- Solve numericals on slip and rotor frequency

Syllabus context

Three-phase induction motor: concept of rotating magnetic field, principle of operation, constructional features of motor, types: squirrel cage and wound rotor.

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Why the induction motor?

- Most widely used AC motor: ~50 % of industrial electric power.
- Robust, simple, low cost, self-starting, little maintenance.
- No electrical connection to the rotor in the squirrel-cage type.
- Constant-speed type: speed falls only 2–5 % from no load to full load.

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Construction

Stator

Hollow cylinder of laminated silicon steel with slots; balanced three-phase winding in star or delta; frame.

Rotor

Laminated cylindrical core on the shaft with slots; squirrel-cage bars or three-phase winding.

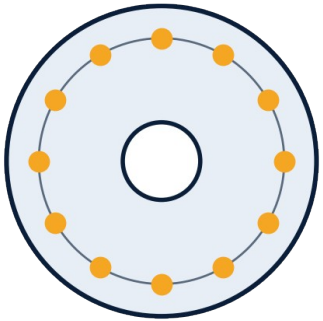
Air gap

Very small (0.3–1 mm) to keep magnetising current low.

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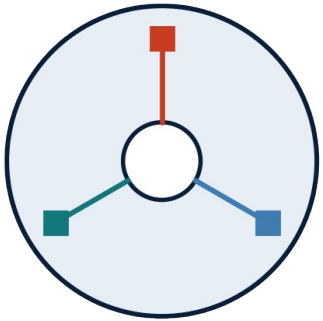
Rotor types

Squirrel-cage rotor



bars + end rings (short-circuited)

Wound (slip-ring) rotor



3 slip rings + brushes → external R

- Squirrel cage: aluminium/copper bars short-circuited by end rings.
- Wound rotor: three-phase winding (star) brought out to slip rings; external resistance can be added.

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Squirrel cage versus slip ring

Feature	Squirrel cage	Slip ring (wound rotor)
Rotor	Bars + end rings	3-phase winding, slip rings, brushes
External resistance	Not possible	Possible: high starting torque
Starting torque	Low–moderate	High
Cost / maintenance	Low / very little	Higher / brushes
Applications	Pumps, fans, machine tools	Cranes, hoists, crushers

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Principle of operation

1

Rotating field

Stator currents produce a field of 1.5 φm rotating at Ns.

2

EMF induced

The field cuts rotor conductors: Faraday's law induces an EMF.

3

Rotor current

Short-circuited rotor carries current; by Lenz's law it opposes the relative motion.

4

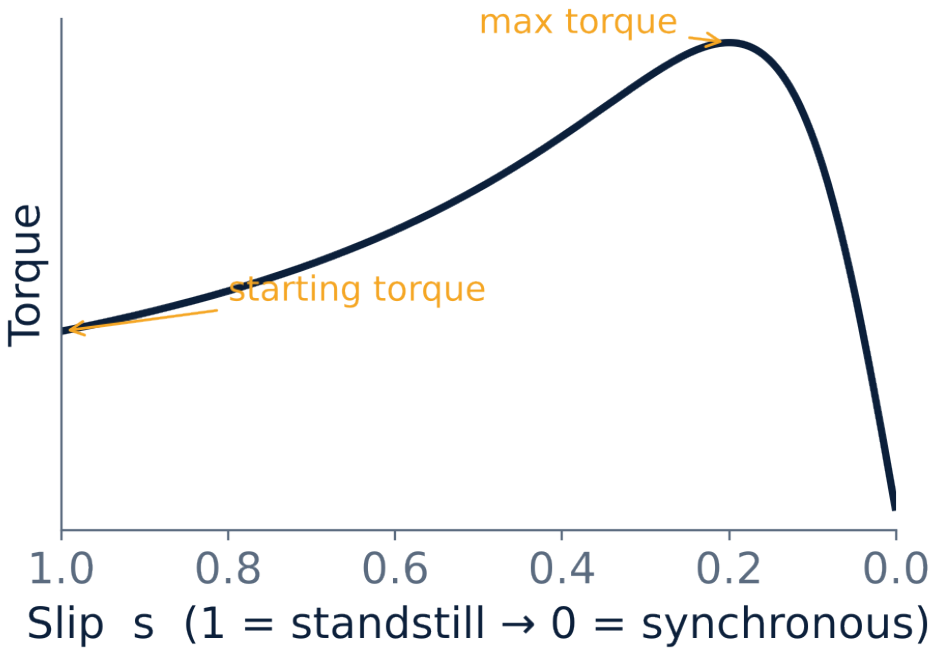
Torque

Rotor is dragged in the direction of the field but never reaches Ns.

Slip s = (Ns − N)/Ns. Rotor speed N = Ns (1 − s). Rotor frequency fr = s f.

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Torque–slip characteristic



- Starting torque at s = 1.
- Maximum torque at about s = 0.15–0.25.
- Full-load slip 2–5 %: stable operation on the right.

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Numerical: slip and rotor frequency

Problem A 4-pole, 50 Hz induction motor runs at 1440 rpm. Find Ns, slip, and rotor frequency.

- $N_s = 120\,f/P = 120 \times 50/4 = 1500$ rpm
- $s = (N_s - N)/N_s = (1500 - 1440)/1500 = 0.04$
- $f_r = s\,f = 0.04 \times 50$

Answer: N_s = 1500 rpm, s = 4 %, f_r = 2 Hz

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Key takeaways

- Stator: 3-phase winding; rotor: squirrel cage or wound
- Rotating field induces rotor current; torque follows Lenz's law
- Slip s = (Ns − N)/Ns; N = Ns(1 − s); fr = s f
- Wound rotor allows added resistance for high starting torque

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Practice questions

Q1. Compare squirrel-cage and slip-ring induction motors.

Q2. A 6-pole, 50 Hz motor runs at 960 rpm. Find slip.

Q3. Why can the rotor never reach synchronous speed?

Q4. What is the rotor frequency at standstill?